Fisher Forecast for the Stellar–Halo Mass Relation in the Dwarf Galaxy Regime with Spec-S5

SHMR Fisher Forecast Pipeline

March 2026

Abstract

We present a Fisher matrix forecast assessing how well current and next-generation spectroscopic surveys, combined with Stage-IV imaging weak lensing, can constrain the stellar–halo mass relation (SHMR) in the dwarf galaxy regime ($M_* < 10^9\,M_\odot$). Four survey scenarios spanning DESI through a futuristic Stage-V "Dwarf Mania" concept are compared in two configurations: a dwarf-only forecast ($\log M_* \in [7, 9]$) and a full stellar mass range forecast ($\log M_* \in [7, 12]$). We find that the low-mass slope β_0 and normalization N_0 of the SHMR can be constrained to sub-percent precision by Spec-S5 and Stage-V surveys, while the mass-dependent scatter parameters ($\sigma_{\rm high}$, $\sigma_{\rm rise}$) remain poorly constrained ($> 300\%$ fractional error) unless the sample spans the full stellar mass range across both sides of the SHMR peak at $M_{\rm h} \sim 10^{12}\,M_\odot$. With full-range coverage, scatter constraints improve by $\sim$100$\times$, reaching $\sim$1% precision on $\sigma_{\rm high}$ for Stage-V.

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1 Executive Summary

1.1 Science Motivation

The stellar–halo mass relation (SHMR) is the central empirical mapping between dark matter halos and the galaxies they host. While the SHMR is well-constrained around $M_{\rm h} \sim 10^{12}\,M_\odot$ (the Milky Way regime), the dwarf galaxy regime ($M_* < 10^{9.5}\,M_\odot$, $M_{\rm h} < 10^{11.5}\,M_\odot$) remains poorly characterized due to the difficulty of obtaining spectroscopic redshifts for faint, low-surface-brightness galaxies. Spec-S5 and other next-generation spectroscopic surveys will dramatically increase the number of dwarf galaxies with spectroscopic redshifts, enabling precise galaxy–galaxy lensing and clustering measurements that directly probe the SHMR at low masses.

1.2 Forecast Pipeline Overview

The forecast pipeline computes the Fisher information matrix from three observables per (redshift, stellar mass) bin:

1. **Galaxy–galaxy lensing** $\Delta\Sigma(R)$: the excess surface mass density around lens galaxies, computed from an HOD-weighted NFW 1-halo profile plus an analytic linear-bias 2-halo term.

2. **Effective bias** $b_{\rm eff}$: the HOD-weighted mean linear halo bias, probing halo mass through large-scale clustering.

3. **Stellar mass function** Φ: the galaxy number density per stellar mass bin, with a covariance model that includes both Poisson noise and cosmic variance.

Key pipeline features:

- **SHMR model:** Moster et al. (2013) double power-law with linear $z/(1+z)$ evolution (8 shape parameters).

- **Mass-dependent scatter:** Cao & Tinker (2020) tanh parameterization with $\sigma(M_{\rm h})$ varying from 0.18 dex at high $M_{\rm h}$ to 0.38 dex at low $M_{\rm h}$.

- **Stellar mass uncertainty:** $\sigma_{\rm obs} = 0.1$ dex for spectroscopic surveys, incorporated as $\sigma_{\rm eff} = \sqrt{\sigma^2_{\rm SHMR} + \sigma^2_{\rm obs}}$.

- **Nuisance parameters:** multiplicative shear calibration bias ($\sigma_m = 0.02$) and source photo-z bias ($\sigma_{\delta z} = 0.03$), marginalized with Gaussian priors.

- **Systematic floor:** 5% irreducible systematic uncertainty on $\Delta\Sigma$ from baryonic effects, miscentering, and photometric calibration.

- **Fixed imaging survey:** Stage-IV lensing ($n_{\rm source} = 25\,{\rm arcmin}^{-2}$, $\sigma_e = 0.26$, $z_{\rm source,med} = 1.0$).

1.3 Key Conclusions

1. **Shape parameters are well-constrained by dwarfs alone.** Even the most limited survey (DESI, 10^5 galaxies) constrains the low-mass slope β_0 to 3.3% and the normalization N_0 to 17%. Spec-S5 (10^7 galaxies, $\log M_{*\mathrm{min}} = 7.0$) achieves sub-percent precision on β_0 (1.0%) and 4.4% on N_0.

2. **Scatter constraints require the full stellar mass range.** In the dwarf-only configuration, even the Stage-V survey cannot constrain σ_{high} or σ_{rise} to better than $\sim$300% fractional error. This is because all dwarf halos sit on the same side of the SHMR peak, where scatter primarily smears galaxies within the same regime without creating distinctive signatures across multiple bins.

3. **Full-range coverage transforms scatter constraints.** Extending to $\log M_* \in [7, 12]$ enables $\sim 100\times$ improvement in scatter precision, reaching 0.9% on σ_{high} and 3.6% on σ_{rise} for Stage-V. The critical lever arm comes from spanning both sides of the SHMR peak at $M_{\mathrm{h}} \sim 10^{12}\, M_\odot$.

4. **Spec-S5 delivers 3–6$\times$ improvement over DESI** across all SHMR parameters in the dwarf regime, driven by deeper stellar mass completeness ($\log M_{*\mathrm{min}} = 7.0$ vs. 8.0) and larger galaxy samples.

2 Survey Configurations

We compare four spectroscopic survey scenarios, all combined with the same Stage-IV imaging weak lensing survey. Two forecast configurations are run for each survey: (i) a *dwarf-only* configuration restricted to $\log M_* < 9.0$ with the high-mass SHMR parameters (γ_0, $\log M_{1,0}$) fixed, and (ii) a *full-range* configuration spanning $\log M_* \in [7, 12]$ with all SHMR shape and scatter parameters freed and galaxy counts scaled by 3$\times$ to reflect the larger samples available when the full stellar mass range is included.

2.1 Dwarf-Only Configuration

Table 1: Survey parameters for the dwarf-only forecast. All surveys use $\Delta \log M_* = 0.5\,\mathrm{dex}$ bins and fix $\gamma_0 = 0.608$ and $\log M_{1,0} = 11.59$. The varied SHMR parameters are N_0, β_0, σ_{high}, σ_{rise}, plus two nuisance parameters (shear calibration m and source photo-z bias δz_s).

Survey	Area [deg^2]	z-range	N_{gal}	$\log M_{*\mathrm{min}}$ [dex]	$\log M_{*\mathrm{max}}$ [dex]	$N_{\mathrm{M*\text{-bins}}}$
DESI	14 000	0.01–0.08	10^5	8.0	9.0	2
DESI-II	5 000	0.01–0.15	10^6	7.5	9.0	3
Spec-S5	5 000	0.01–0.20	1.2×10^7	7.0	9.0	4
Stage-V DM	8 000	0.01–0.30	4×10^7	7.0	9.0	4

2.2 Full Stellar Mass Range Configuration

Table 2: Survey parameters for the full-range forecast. The upper stellar mass limit is extended to $\log M_* = 12.0$, and galaxy counts are scaled by 3× relative to the dwarf-only configuration. All six $z = 0$ SHMR shape parameters are varied: $\log M_{1,0}$, N_0, β_0, γ_0, plus $\sigma_{\rm high}$ and $\sigma_{\rm rise}$.

Survey	Area [deg^2]	z-range	$N_{\rm gal}$	$\log M_{*\rm min}$ [dex]	$\log M_{*\rm max}$ [dex]	$N_{\rm M*\text{-bins}}$
DESI	14 000	0.01–0.08	3×10^5	8.0	12.0	8
DESI-II	5 000	0.01–0.15	3×10^6	7.5	12.0	9
Spec-S5	5 000	0.01–0.20	3.6×10^7	7.0	12.0	10
Stage-V DM	8 000	0.01–0.30	1.2×10^8	7.0	12.0	10

3 Forecast Results: Dwarf-Only

3.1 Lensing Signal and Signal-to-Noise

Figure 1 shows the predicted galaxy–galaxy lensing signal $\Delta\Sigma(R)$ for the lowest common stellar mass bin ($\log M_* \in [8.0, 8.5]$ at $z = 0.05$), with 1σ error bars for each survey. The signal peaks at $\sim 3\, M_\odot/\mathrm{pc}^2$ near $R \sim 0.05\,\mathrm{Mpc}$ and transitions from the 1-halo NFW regime to the 2-halo linear-bias regime at $R \sim 0.3\,\mathrm{Mpc}$. DESI (blue) has the largest error bars due to its small galaxy count (10^5) and narrow redshift range, while Stage-V (red) achieves the smallest errors from its combination of large area, deep completeness, and extended redshift range.

3.2 Parameter Constraints

Figure 2 presents the fractional parameter errors $\sigma(\theta)/|\theta_{\rm fid}|$ for all four surveys. The shape parameters (N_0, β_0) are well-constrained: Spec-S5 achieves 4.4% on N_0 and 1.0% on β_0, while Stage-V reaches 3.1% and 0.7% respectively. In contrast, the scatter parameters ($\sigma_{\rm high}$, $\sigma_{\rm rise}$) have fractional errors exceeding 300% for all surveys, indicating that the dwarf-only sample cannot meaningfully constrain the mass-dependent scatter model.

3.3 Fisher Ellipses and Parameter Degeneracies

Figure 3 shows the 2D Fisher forecast ellipses (1σ and 2σ) for three key parameter pairs. The left panel (N_0 vs. β_0) shows compact, well-oriented ellipses that tighten systematically from DESI to Stage-V. The center panel ($\sigma_{\rm high}$ vs. $\sigma_{\rm rise}$) reveals a tight degeneracy: the dwarf-only data constrains a *combination* of these parameters (the effective scatter at the characteristic dwarf halo mass) but cannot disentangle the plateau height from the transition steepness. The right panel (N_0 vs. $\sigma_{\rm high}$) shows a mild correlation between the SHMR normalization and the scatter amplitude.

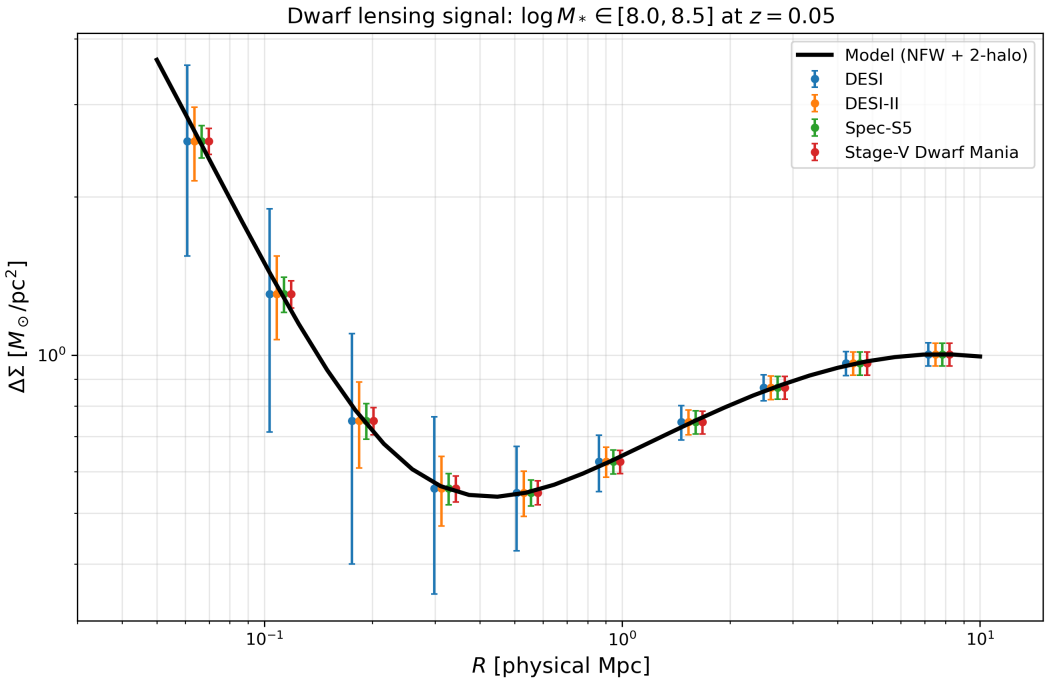

Figure 1: Galaxy–galaxy lensing signal $\Delta\Sigma(R)$ in the dwarf stellar mass bin $\log M_* \in [8.0, 8.5]$ at $z = 0.05$, with 1σ error bars for each survey. The solid black curve shows the model prediction (NFW 1-halo + linear-bias 2-halo). Error bars include shape noise from Stage-IV imaging lensing ($n_{\rm source} = 25\,{\rm arcmin}^{-2}$, $\sigma_e = 0.26$) plus a 5% systematic floor. Radial bins span $R \in [0.01, 10]$ Mpc with 30 log-spaced bins, emphasizing the 1-halo regime where most Fisher information resides.

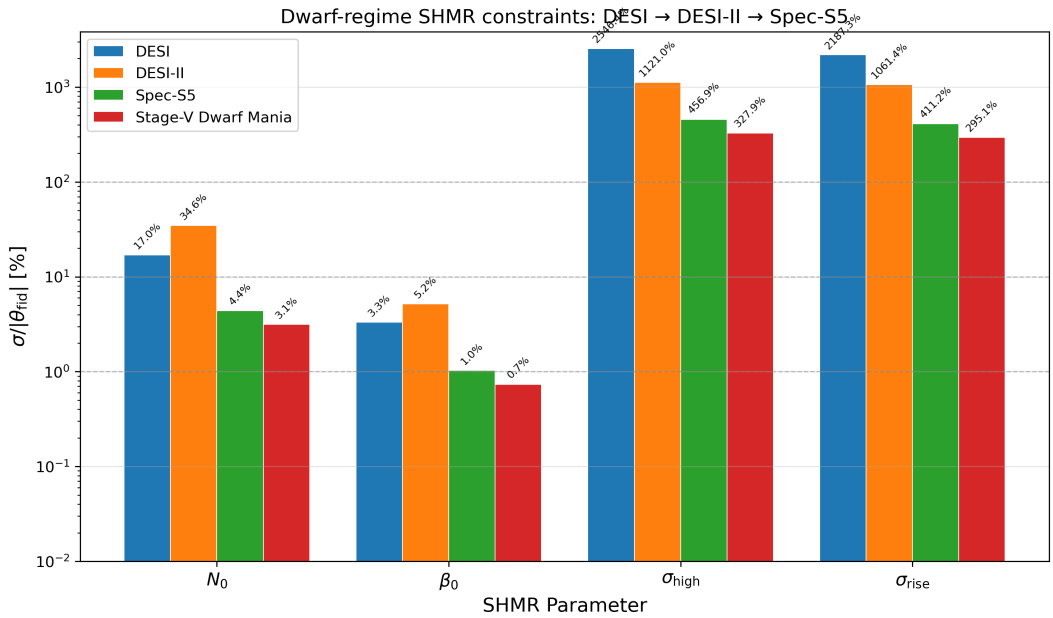

Figure 2: Fractional 1σ parameter errors (log scale) for the dwarf-only forecast. The SHMR shape parameters (N_0, β_0) are constrained to a few percent, with systematic improvements from DESI through Stage-V. The scatter parameters ($\sigma_{\rm high}$, $\sigma_{\rm rise}$) remain unconstrained ($> 300\%$) because all dwarf halos probe the same side of the SHMR peak. Reference lines at 1%, 10%, and 100% are shown.

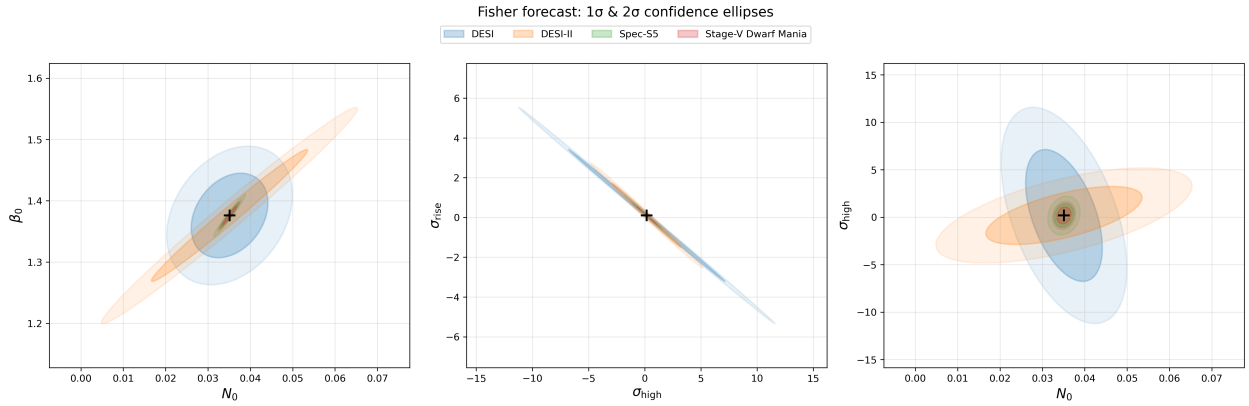

Figure 3: Fisher forecast 1σ (dark shading) and 2σ (light shading) confidence ellipses for three parameter pairs in the dwarf-only configuration. *Left:* N_0 vs. β_0 — compact ellipses showing these shape parameters are well-constrained. *Center:* $\sigma_{\rm high}$ vs. $\sigma_{\rm rise}$ — a strong degeneracy along a narrow axis indicates that dwarf-only data constrains only a linear combination of the scatter parameters. *Right:* N_0 vs. $\sigma_{\rm high}$ — mild correlation between normalization and scatter. The fiducial parameter values are marked with a cross.

3.4 Relative Improvement over DESI

Figure 4 quantifies the improvement factor $\sigma(\mathrm{DESI})/\sigma(\mathrm{survey})$ for each survey relative to DESI. Spec-S5 achieves 3.2–5.6× improvement across all parameters, driven by its deeper stellar mass completeness ($\log M_{*\mathrm{min}} = 7.0$ vs. 8.0, adding two additional low-mass bins) and 120× more galaxies. Stage-V provides a further factor of ~ 1.4 beyond Spec-S5, reaching 4.5–7.8× improvement over DESI. DESI-II is comparable to or slightly worse than DESI on shape parameters (smaller area partially offsets the deeper sample), but $\sim$2× better on scatter parameters due to the additional low-mass bin at $\log M_* = 7.5$.

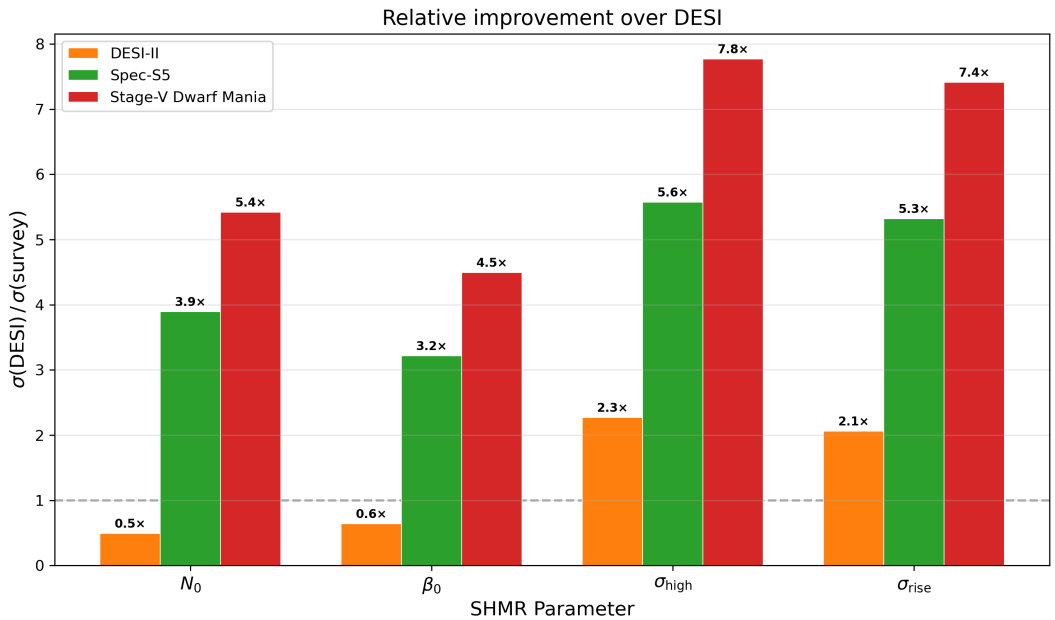

Figure 4: Improvement factor $\sigma(\mathrm{DESI})/\sigma(\mathrm{survey})$ for each SHMR parameter relative to the DESI baseline. Values above 1 indicate better constraints than DESI. Spec-S5 (green) achieves 3–6× improvement; Stage-V Dwarf Mania (red) reaches 4.5–8×. DESI-II (orange) is slightly worse than DESI on N_0 and β_0 due to its smaller area (5 000 vs. 14 000 deg^2) but better on scatter parameters from the additional low-mass bin.

4 Forecast Results: Full Stellar Mass Range

Extending the stellar mass range to $\log M_* \in [7, 12]$ transforms the forecast by spanning both sides of the SHMR peak at $M_\mathrm{h} \sim 10^{12}\,M_\odot$. With the full range, six SHMR shape parameters are varied ($\log M_{1,0}$, N_0, β_0, γ_0, σ_high, σ_rise), and galaxy counts are scaled by 3×.

4.1 Parameter Constraints

Figure 5 shows fractional errors for the full-range configuration. All parameters are now constrained to $< 12\%$ even for DESI; Stage-V achieves sub-percent precision on all shape parameters and 0.9% on σ_high. The scatter parameter σ_rise, which controls the steepness of

the scatter transition near $M_{\rm h} \sim 10^{12}\,M_\odot$, is constrained to 3.6% by Stage-V (compared to 295% in the dwarf-only case — a $\sim$80$\times$ improvement).

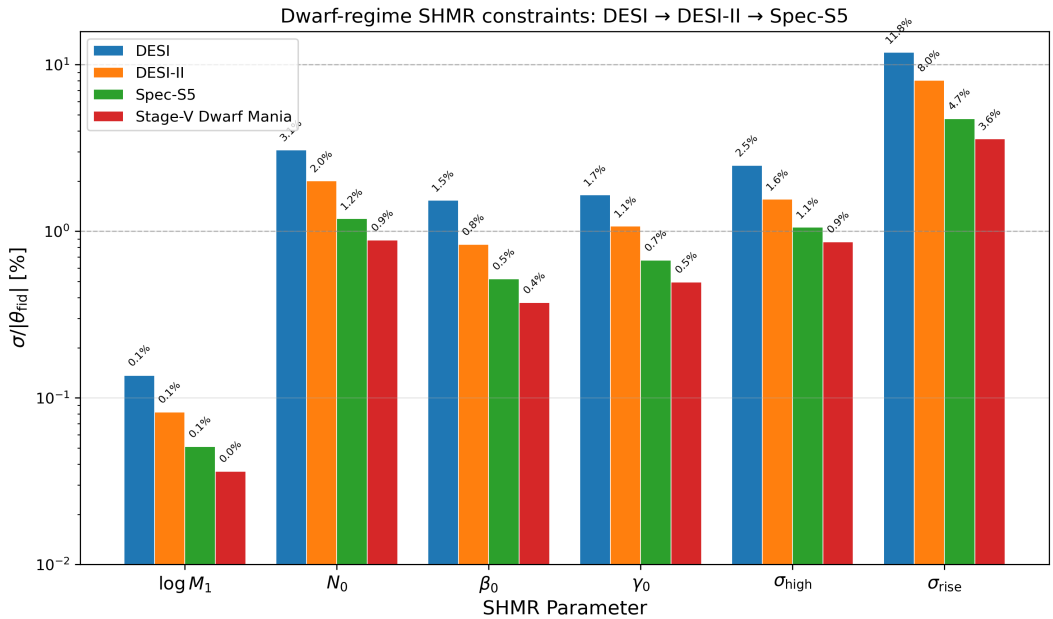

Figure 5: Fractional 1σ errors for the full stellar mass range forecast ($\log M_* \in [7, 12]$). All six SHMR parameters are now well-constrained ($< 12\%$). Stage-V reaches sub-percent precision on all shape parameters, with $\sigma_{\rm high}$ constrained to 0.9% — a $\sim$360$\times$ improvement over the dwarf-only forecast.

4.2 Fisher Ellipses

Figure 6 shows the Fisher ellipses for the full-range case. The $\sigma_{\rm high}$–$\sigma_{\rm rise}$ degeneracy (center panel) is dramatically reduced compared to the dwarf-only case (Figure 3): the ellipses are compact rather than razor-thin, indicating that the full mass range breaks the degeneracy between the scatter plateau and transition steepness. The physical mechanism is that high-mass bins ($M_* > 10^{10.5}$) probe halos where $\sigma \to \sigma_{\rm high}$, while dwarf bins probe halos where $\sigma \to \sigma_{\rm high} + 2\sigma_{\rm rise}$; together they separate the two parameters.

4.3 Relative Improvement

Figure 7 shows the improvement factors for the full-range case. The gains are more uniform across parameters compared to the dwarf-only case: Stage-V achieves 2.9–4.1$\times$ improvement over DESI on all six parameters, with the largest gain on β_0 (4.1$\times$) driven by the low-mass bins that sensitively probe the SHMR slope. Spec-S5 achieves 2.4–3.0$\times$ improvement, establishing it as the primary science driver for constraining the SHMR at the next-generation level.

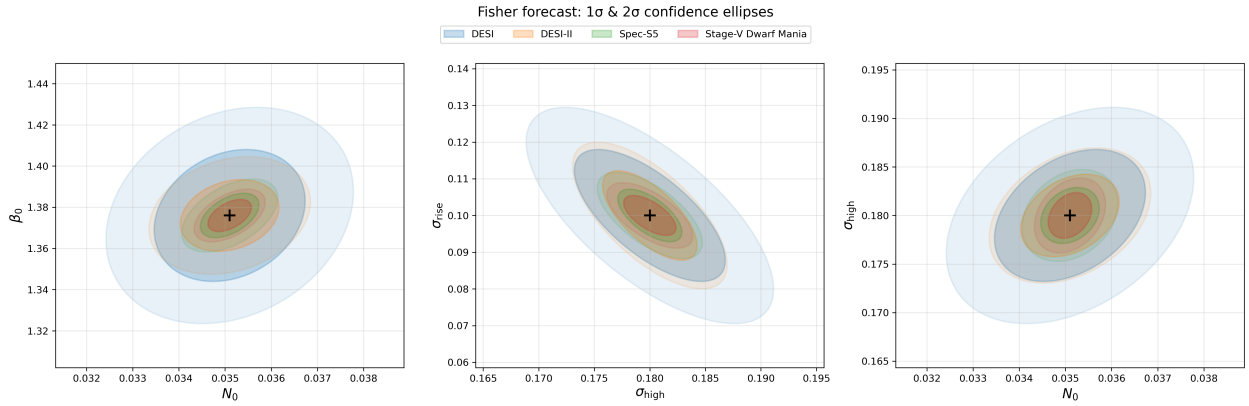

Figure 6: Fisher ellipses for the full stellar mass range forecast. Compared to the dwarf-only case (Figure 3), the $\sigma_{\rm high}$–$\sigma_{\rm rise}$ degeneracy (center) is broken by spanning both sides of the SHMR peak. All parameter pairs show compact, well-separated ellipses, with Stage-V (innermost) achieving $\sim$3$\times$ tighter constraints than DESI (outermost).

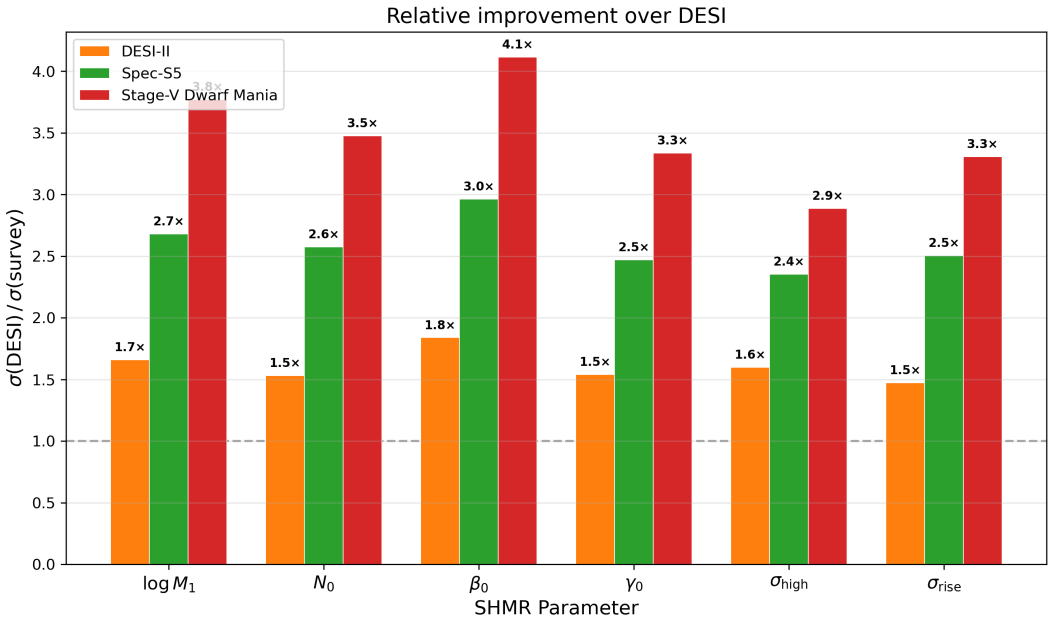

Figure 7: Improvement factor over DESI for the full stellar mass range. The gains are more uniform across all six parameters compared to the dwarf-only case, reflecting the complementary information from high-mass and low-mass bins. Stage-V achieves 2.9–4.1$\times$ improvement; Spec-S5 achieves 2.4–3.0$\times$.

5 Technical Details

5.1 SHMR Parameterization

The mean SHMR follows the Moster et al. (2013) double power-law:

$$\frac{M_*}{M_\mathrm{h}} = \frac{2\,N(z)}{(M_\mathrm{h}/M_1(z))^{-\beta(z)} + (M_\mathrm{h}/M_1(z))^{\gamma(z)}}, \tag{1}$$

where each shape parameter evolves linearly in the variable $\zeta \equiv z/(1+z)$:

$$X(z) = X_0 + \nu_X \cdot \zeta. \tag{2}$$

Fiducial values are from Moster et al. (2013) Table 1: $\log M_{1,0} = 11.59$, $N_0 = 0.0351$, $\beta_0 = 1.376$, $\gamma_0 = 0.608$. The inverse relation $M_\mathrm{h}(M_*)$ is obtained via `scipy.optimize.brentq` root-finding, exploiting the monotonicity of the SHMR.

5.1.1 Mass-Dependent Scatter

At fixed halo mass, the stellar mass is log-normally distributed with scatter following Cao & Tinker (2020):

$$\sigma(M_\mathrm{h}) = \sigma_\mathrm{high} + \sigma_\mathrm{rise}\left[1 - \tanh\left(\frac{\log M_\mathrm{h} - \log M_\mathrm{h}^\mathrm{break}}{\delta}\right)\right], \tag{3}$$

with fiducial values $\sigma_\mathrm{high} = 0.18\,\mathrm{dex}$, $\sigma_\mathrm{rise} = 0.10\,\mathrm{dex}$, $\log M_\mathrm{h}^\mathrm{break} = 12.0$, $\delta = 0.4\,\mathrm{dex}$. The asymptotic scatter is 0.18 dex at high M_h (cluster-scale halos) and 0.38 dex at low M_h (dwarf halos). Only σ_high and σ_rise are varied in the Fisher matrix; the transition mass and width are held fixed.

5.2 Halo Occupation Distribution

The continuous SHMR with scatter is converted into a halo occupation distribution (HOD) for each stellar mass bin $[\log M_*^\mathrm{lo}, \log M_*^\mathrm{hi}]$:

$$\langle N_\mathrm{cen}(M_\mathrm{h})\rangle = \frac{1}{2}\left[\mathrm{erf}(x_\mathrm{hi}) - \mathrm{erf}(x_\mathrm{lo})\right], \quad x = \frac{\log M_*^\mathrm{edge} - \langle \log M_*\rangle(M_\mathrm{h})}{\sqrt{2}\,\sigma_\mathrm{eff}(M_\mathrm{h})}, \tag{4}$$

where $\sigma_\mathrm{eff}(M_\mathrm{h}) = \sqrt{\sigma^2_\mathrm{SHMR}(M_\mathrm{h}) + \sigma^2_\mathrm{obs}}$ is the effective scatter incorporating stellar mass measurement uncertainty. For spectroscopic surveys, we adopt $\sigma_\mathrm{obs} = 0.1\,\mathrm{dex}$.

Satellites follow Leauthaud et al. (2012):

$$\langle N_\mathrm{sat}(M_\mathrm{h})\rangle = \langle N_\mathrm{cen}^\mathrm{thresh}\rangle \cdot \left(\frac{M_\mathrm{h}}{M_{1,\mathrm{sat}}}\right)^{\alpha_\mathrm{sat}}, \tag{5}$$

with $M_{1,\mathrm{sat}} = 17\,M_\mathrm{min}$ and $\alpha_\mathrm{sat} = 1.0$. Satellite parameters are fixed (not varied in the Fisher matrix); this is acceptable for relative survey comparisons.

5.3 Observable Predictions

5.3.1 Galaxy–Galaxy Lensing

The bin-averaged excess surface mass density is:

$$\Delta\Sigma(R\,|\,\mathrm{bin}, z) = \frac{\int dM_\mathrm{h} \frac{dn}{dM_\mathrm{h}}\, N_\mathrm{tot}(M_\mathrm{h})\, [\Delta\Sigma_\mathrm{NFW}(R, M_\mathrm{h}, z) + b_h(M_\mathrm{h}, z)\, \Delta\Sigma_\mathrm{mm}(R, z)]}{n_\mathrm{gal}}, \tag{6}$$

where dn/dM_h is the Tinker et al. (2008) halo mass function, $\Delta\Sigma_\mathrm{NFW}$ uses the Wright & Brainerd (2000) analytic expression with Diemer & Kravtsov (2019) concentrations and the `200m` mass definition, and b_h is the Tinker et al. (2010) linear halo bias. The 2-halo matter term $\Delta\Sigma_\mathrm{mm}(R, z)$ is computed from the nonlinear matter correlation function via line-of-sight projection (truncated at $\pi_\mathrm{max} = 100\,\mathrm{Mpc}$) and cached per redshift.

All integrals use the trapezoidal rule on a log-spaced grid of 200 halo masses spanning $\log M_\mathrm{h} \in [10.0, 15.5]$. Colossus NFW profiles work in comoving kpc/h; the conversion to physical $M_\odot/\mathrm{Mpc}^2$ is:

$$\Delta\Sigma_\mathrm{phys} = \Delta\Sigma_\mathrm{colossus} \times h \times 10^6 \times (1+z)^2. \tag{7}$$

5.3.2 Effective Bias

$$b_\mathrm{eff}(\mathrm{bin}, z) = \frac{\int dM_\mathrm{h} \frac{dn}{dM_\mathrm{h}}\, N_\mathrm{tot}(M_\mathrm{h})\, b_h(M_\mathrm{h}, z)}{n_\mathrm{gal}}. \tag{8}$$

5.3.3 Stellar Mass Function

$$\Phi(\mathrm{bin}, z) = n_\mathrm{gal} = \int dM_\mathrm{h} \frac{dn}{dM_\mathrm{h}}\, N_\mathrm{tot}(M_\mathrm{h}) \quad [\mathrm{Mpc}^{-3}]. \tag{9}$$

5.4 Covariance Models

5.4.1 Lensing Covariance

The lensing covariance is diagonal (no off-diagonal radial bin correlations) and shape-noise dominated:

$$\mathrm{Var}(\Delta\Sigma_i) = \frac{\sigma_e^2\, \Sigma_\mathrm{crit,eff}^2}{N_\mathrm{lens} \cdot n_\mathrm{source,eff} \cdot A_\mathrm{ann,i}} + (f_\mathrm{sys}\, \Delta\Sigma_\mathrm{fid})^2\,, \tag{10}$$

where $A_\mathrm{ann,i} = \pi(R_\mathrm{hi}^2 - R_\mathrm{lo}^2)$ is the annulus area in physical Mpc^2, $\Sigma_\mathrm{crit,eff}$ is the effective critical surface mass density averaged over a Smail-type source redshift distribution $n(z_s) \propto z_s^2\, \exp[-(z_s/z_0)^{1.5}]$, and $n_\mathrm{source,eff}$ is the effective source density behind the lens redshift. The systematic floor $f_\mathrm{sys} = 0.05$ accounts for baryonic effects, miscentering, and photometric calibration.

5.4.2 Clustering Covariance

The effective bias variance includes cosmic variance and shot noise via the FKP-weighted effective volume:

$$\mathrm{Var}(b_{\mathrm{eff}}) = \frac{b_{\mathrm{eff}}^2}{n_{\mathrm{gal}} \cdot V_{\mathrm{eff}}}, \quad V_{\mathrm{eff}} = V_{\mathrm{survey}} \cdot \left[\frac{nPb^2}{1+nPb^2}\right]^2, \tag{11}$$

where P is evaluated at $k = 0.1\,h/\mathrm{Mpc}$.

5.4.3 SMF Covariance

The SMF covariance includes Poisson noise and cosmic variance:

$$\mathrm{Var}(\Phi) = \frac{\Phi}{V_{\mathrm{survey}}} + b_{\mathrm{eff}}^2\,\sigma_m^2(V,z)\,\Phi^2, \tag{12}$$

where $\sigma_m^2(V,z)$ is the matter variance on the survey scale, computed from the colossus matter power spectrum with a top-hat window function of volume V. In the Poisson-dominated regime (small V or rare populations), $\sigma(\Phi)/\Phi \approx 1/\sqrt{N}$. In the cosmic-variance-dominated regime (large V, abundant populations), $\sigma(\Phi)/\Phi \approx b_{\mathrm{eff}}\,\sigma_m$, which sets an irreducible floor that does not decrease with more galaxies.

5.5 Fisher Matrix Assembly

5.5.1 Parameter Selection

The varied parameter set depends on the forecast configuration:

- **Dwarf-only:** N_0, β_0, σ_{high}, σ_{rise} (4 SHMR parameters), plus shear calibration m and source photo-z bias δz_s (2 nuisance, with Gaussian priors $\sigma_m = 0.02$ and $\sigma_{\delta z} = 0.03$). The high-mass parameters γ_0 and $\log M_{1,0}$ are fixed because dwarf halos ($M_{\mathrm{h}} \ll M_1 \sim 10^{11.6}\,M_\odot$) are insensitive to them.
- **Full-range:** $\log M_{1,0}$, N_0, β_0, γ_0, σ_{high}, σ_{rise} (6 SHMR parameters), plus the same 2 nuisance parameters.
- **No z-evolution:** Since all surveys have $z_{\mathrm{max}} \leq 0.3$, the evolution parameters (ν_{M_1}, ν_N, ν_β, ν_γ) are not varied.

5.5.2 Numerical Derivatives

Derivatives of observables with respect to SHMR parameters are computed via central finite differences:

$$\frac{\partial f}{\partial \theta_i} \approx \frac{f(\theta_i+\delta) - f(\theta_i-\delta)}{2\delta}, \quad \delta = 0.01 \times |\theta_i|. \tag{13}$$

For parameters near zero ($|\theta_i| < 10^{-10}$), an absolute step of 0.01 is used. Nuisance parameter derivatives are analytic.

5.5.3 Fisher Accumulation

For each (redshift bin, stellar mass bin), the Fisher matrix receives three additive contributions:

$$F = \sum_{\text{bins}} \left[J_\Delta \Sigma^\top C_\Delta \Sigma^{-1} J_\Delta \Sigma + \frac{J_b J_b^\top}{\text{Var}(b_{\text{eff}})} + \frac{J_\Phi J_\Phi^\top}{\text{Var}(\Phi)} \right], \tag{14}$$

where $J_\Delta \Sigma$, J_b, J_Φ are the Jacobian matrices of the lensing, bias, and SMF observables, and $C_\Delta \Sigma$ is the diagonal lensing covariance. Gaussian priors on nuisance parameters are added to the Fisher diagonal after accumulation.

5.5.4 Post-Processing

Marginalized 1σ errors: $\sigma(\theta_i) = \sqrt{(F^{-1})_{ii}}$. Fisher ellipses for parameter pairs (i, j) are computed from the 2×2 sub-covariance of F^{-1} via eigendecomposition. The 1σ ellipse corresponds to $\Delta\chi^2 = 2.30$ (68.3% CL for 2 parameters), and the 2σ ellipse to $\Delta\chi^2 = 6.18$ (95.4% CL). The condition number of F is checked; values $> 10^{10}$ trigger a warning.

5.6 Stellar Mass Measurement Uncertainty

Observed stellar masses carry a statistical uncertainty $\sigma_{\text{obs}} \approx 0.1\,\text{dex}$ for spectroscopic surveys. This is incorporated exactly by replacing the intrinsic scatter $\sigma_{\text{SHMR}}(M_{\text{h}})$ with the effective scatter $\sigma_{\text{eff}}(M_{\text{h}}) = \sqrt{\sigma^2_{\text{SHMR}}(M_{\text{h}}) + \sigma^2_{\text{obs}}}$ in the HOD computation (Eq. 4). This is the exact convolution of two Gaussians in log-space; no numerical integration is needed.

The impact is largest at high halo masses where σ_{SHMR} is smallest (0.18 dex, giving +15% increase in effective scatter). For dwarf galaxies ($\sigma_{\text{SHMR}} \approx 0.38\,\text{dex}$), the effect is negligible (+3% in quadrature).

5.7 Binning and Integration

- **Redshift bins:** edges from z_{min} to z_{max} in steps of $\Delta z = 0.2$; each bin evaluated at z_{mid}. For the dwarf surveys ($z_{\text{max}} \leq 0.3$), this yields a single redshift bin.
- **Stellar mass bins:** from $\log M_{*\text{min}}$ to $\log M_{*\text{max}}$ in steps of 0.5 dex.
- **Radial bins for lensing:** 30 log-spaced bins from $R_{\text{min}} = 0.01$ to $R_{\text{max}} = 10\,\text{Mpc}$.
- **Halo mass grid:** 200 log-spaced points from $\log M_{\text{h}} = 10.0$ to 15.5.

5.8 Cosmology

All calculations adopt Planck 2018 cosmology (Planck Collaboration 2020) via `colossus`: $h = 0.6736$, $\Omega_m = 0.3153$, $\Omega_b = 0.0493$, $\sigma_8 = 0.8111$, $n_s = 0.9649$.

6 Key Assumptions and Limitations

1. **NFW + linear 2-halo term.** The 1-halo–2-halo transition region ($R \sim 1$–$3\,\mathrm{Mpc}$) is not explicitly modeled; the two terms are summed directly. This double-counts matter near the virial radius but has negligible impact on the Fisher matrix, which depends on parameter derivatives rather than absolute signal levels.

2. **Diagonal lensing covariance.** Off-diagonal correlations between radial bins from large-scale structure are not included, leading to mildly optimistic constraints.

3. **Clustering compressed to $b_{\rm eff}$ + SMF.** The full $w_p(r_p)$ information content is compressed to a single bias value per bin, discarding scale-dependent clustering information.

4. **Fixed satellite prescription.** Satellite parameters ($\alpha_{\rm sat}$, $M_1/M_{\rm min}$) are not varied; absolute error bars should be interpreted with this caveat.

5. **No scatter redshift evolution.** $\sigma(M_{\rm h})$ is assumed constant with z, consistent with Behroozi et al. (2019).

6. **No covariance between stellar mass bins.** Each (z, M_*) bin contributes independently. Overlapping halo populations create inter-bin correlations that are not modeled.

7. **Unbiased stellar mass estimator.** Only statistical scatter ($\sigma_{\rm obs}$) is included; systematic biases in stellar mass estimation are not modeled.

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